

(a) Dashboard navigation

LipidLab
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 DISCOVERY

Explore results

Run a campaign

Published measurements
Inspect saved explanations or evaluate your own measured pairs in a local active-learning campaign.
No synthetic affinities.
No remote upload service.

BENCHMARKS & EXPLAINABILITY
Compare how the model learns.
 Explore the full representation, kernel and acquisition grid. Compare discovery with prediction, then inspect available explanations.

Saved, reproducible experiments

New: leakage-controlled study
 11,016 additional unique campaigns: cold-sequence/scaffold splits, ESM-2 protein embeddings, prediction-aware Thompson and lipid adaptation.
 Explore all new runs Updated full paper SHAP explanations Highlighted structures
 New results use 120 acquired labels plus validation costs; older pooled runs below use 360. Public hosting is read-only.

Choose an experiment
 Analysis: Full benchmark Dataset / target: biodolphin Kd Representation: ChemBERTa-MTR · 384D
 Kernel: linear Acquisition policy: Expected improvement (EI) Seed: 7
 5850 completed benchmark runs · 90 explanation reports

Supported suite: 5 lipid representations × 5 kernels × 13 acquisition policies. Select "Full benchmark" to see every completed combination. Explanation views contain only separately computed SHAP reports.

0.730
Held-out MAE · pK

0.571
Held-out R²

71.4%
Top-decile pool recall

360
Labels · seed 7

Learning curves · selected seed
 (a) Held-out MAE (b) Top-decile discovery

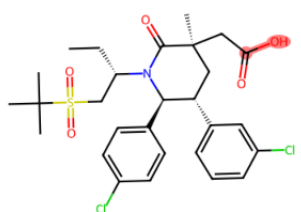
(b) Highlighted structures

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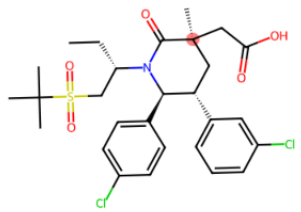
Explore results

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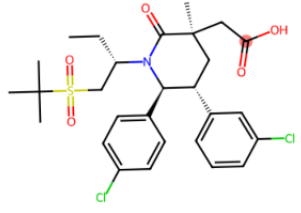
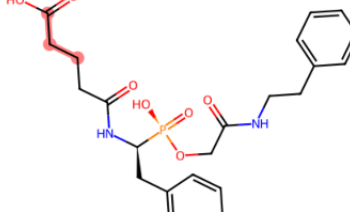
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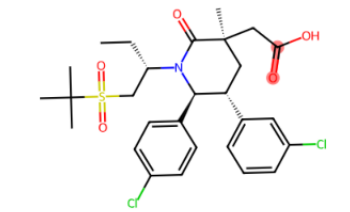
Bit 2437 | pK 10.39
246/864 pairs



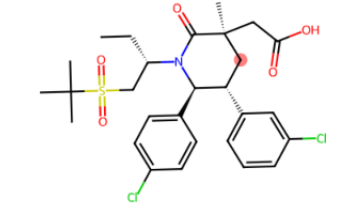
Bit 3108 | pK 10.39
412/864 pairs

Bit 3612 | pK 9.21
284/864 pairs



Bit 1917 | pK 10.39
495/864 pairs



(c) Leakage-controlled explorer

Explore every new configuration
[Original benchmark / SHAP / highlighted structures](#) · [Full paper](#)
 9,720 main campaigns; 1,944 adaptation comparisons, including 648 reused baselines.

This browser exposes every run, including configurations not selected by validation. Do not select a model by its test score. The figures below use validation-only selections.

Study	Endpoint	Evaluation split	Protein representation	Lipid representation
Main: 9,720 campaigns	biodolphin Kd	cold sequence40	esm2 35m	chemberta mlm

Adaptation	Kernel	Acquisition policy	Seed
frozen	linear	prediction aware thompson	7

1.242
Test MAE (pK)

-0.283
Test R²

0.309
Pool recall

0.205
Recall AUC

1.429
Validation MAE

120 acquired labels; 257 total labels including validation. Nonconverged fits: 0. Frozen MLM adaptation entries reuse main-study baselines. No prospective evaluation.

All configurations for the selected study, endpoint and split
[Main per-seed CSV](#) · [Adaptation CSV](#) · [Paired tests](#) · [Split audit](#)

(d) Adaptation explorer

Explore every new configuration
[Original benchmark / SHAP / highlighted structures](#) · [Full paper](#)
 9,720 main campaigns; 1,944 adaptation comparisons, including 648 reused baselines.

This browser exposes every run, including configurations not selected by validation. Do not select a model by its test score. The figures below use validation-only selections.

Study	Endpoint	Evaluation split	Protein representation	Lipid representation
Adaptation: 1,944 comparison	biodolphin Kd	cold sequence40	esm2 35m	chemberta mlm

Adaptation	Kernel	Acquisition policy	Seed
initial label adapter	linear	prediction aware thompson	7

1.242
Test MAE (pK)

-0.283
Test R²

0.309
Pool recall

0.205
Recall AUC

1.429
Validation MAE

120 acquired labels; 257 total labels including validation. Nonconverged fits: 0. Frozen MLM adaptation entries reuse main-study baselines. No prospective evaluation.

All configurations for the selected study, endpoint and split
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